MAT x Library Mathematics Week 2026

Stock Market Game - Data and Formula Conclusion

Prepared for leader review - 30 September 2026

Decision summary

- Confirmed indices: LIB, BOOK, DONA, SCM.
- The current scope contains four confirmed indices. If the leader still requires at least five, a fifth source remains TBC and must not be invented or duplicated from LIB.
- Every index starts at M$100.00.
- Every ranked player starts with M$5,000.00 and zero holdings.
- There are no knowledge points, market points, activity points, or bonus points.
- The leaderboard compares only virtual money: cash plus the latest value of holdings.
- Recommended simplified MVP: no transaction fee, whole units only, long-only, and at most one confirmed action per index per player per day.
- Eight observations produce seven price changes. The eight observations must be prepared and locked before the game.

Daily player cycle

1. The player sees the current price and previously revealed data.
2. The player buys, holds, or sells during the open period.
3. At the daily cutoff, orders are locked.
4. The next preloaded historical observation is revealed.
5. The system calculates the new price.
6. The leaderboard recalculates virtual money.

No player transaction changes the index price. All data and prices are pre-calculated before launch.

Common notation and price rules

- O_t: newly revealed source observation.
- O_(t-1): previous source observation.
- P_t: new index price.
- P_(t-1): previous index price.
- R_t: price return in percent.
- clamp(x, -15, +15): values below -15 become -15; values above +15 become +15.

For LIB, BOOK, and SCM:

$$C_t = 200 \times (O_t - O_{(t-1)}) / (O_t + O_{(t-1)})$$

$$R_t = \text{clamp}(C_t / 5, -15, +15)$$

If both observations are zero, set R_t = 0. Round R_t to one decimal place before updating the price.

For DONA:

$$R_t = \text{clamp}(5 \times \ln((O_t + 1) / (O_{(t-1)} + 1)) / G_t, -15, +15)$$

G_t is the number of calendar months between the two valid observations. The +1 allows a zero month to be processed. The logarithm reduces the effect of unusually large donation batches.

For every index:

$$P_t = \text{round}(P_{(t-1)} \times (1 + R_t / 100), 2)$$

If an observation is missing, closed, or not comparable, hold the previous price and show Data unavailable.

Index decisions and selected data

LIB - Library Footfall Index

Source metric: Gate A + Gate B + Gate C + Gate D for each selected date. The source Total column is not trusted automatically; the four gates are recomputed.

Selected period: 19-26 November 2025, eight consecutive dates.

Observation	Date	Footfall	Return	Price
1	2025-11-19	3,080	Start	M$100.00
2	2025-11-20	2,777	-2.1%	M$97.90
3	2025-11-21	1,931	-7.2%	M$90.85
4	2025-11-22	935	-13.9%	M$78.22
5	2025-11-23	2,863	+15.0%	M$89.95
6	2025-11-24	2,596	-2.0%	M$88.15
7	2025-11-25	2,788	+1.4%	M$89.38
8	2025-11-26	2,830	+0.3%	M$89.65

Data check: the source has 365 dated rows but 364 unique parsed dates, one duplicated date label, and two total mismatches. The selected eight-row window has complete gate values and each recomputed sum matches its reported total. Confirm with the Library that 23 November was a normal operating day.

BOOK - Library Resources Index

Source metric: count of TransactionType = issue for each selected date. renew is excluded, and the file does not contain return transactions.

Selected period: eight recent operating weekdays from 7-18 August 2026. Weekends are skipped, not treated as zero.

Observation	Date	Issues	Return	Price
1	2026-08-07	18	Start	M$100.00
2	2026-08-10	10	-11.4%	M$88.60
3	2026-08-11	5	-13.3%	M$76.82
4	2026-08-12	4	-4.4%	M$73.44
5	2026-08-13	6	+8.0%	M$79.32
6	2026-08-14	5	-3.6%	M$76.46
7	2026-08-17	9	+11.4%	M$85.18
8	2026-08-18	6	-8.0%	M$78.37

Data check: 21,943 transaction rows consist of 18,196 issues and 3,747 renewals. The selected window has no missing observations. Confirm that these dates represent normal Library operation and not a special break.

DONA - Collection Growth Index

Source metric: monthly sum of Books_Donated for records with Status = OK. No donor names or descriptions enter the game data.

Selected observations: January, February, April, May, June, July, August, and September 2026. March is absent and is skipped, not replaced by zero.

Observation	Month	Approved books	Gap	Return	Price
1	2026-01	453	-	Start	M$100.00
2	2026-02	50	1	-10.9%	M$89.10
3	2026-04	74	2	+1.0%	M$89.99
4	2026-05	40	1	-3.0%	M$87.29
5	2026-06	173	1	+7.2%	M$93.57
6	2026-07	816	1	+7.7%	M$100.77
7	2026-08	44	1	-14.5%	M$86.16
8	2026-09	25	1	-2.7%	M$83.83

Data check: 818 source rows include 791 OK, 22 MISSING BOOKS & DATE, three MISSING DATE, and two SUSPICIOUS VALUE records. The two suspicious values are excluded. Confirm that July's total of 816 is genuine and that September was complete when exported.

SCM - Library Social Interaction Index

Recommended source metric: total votes received by one Library Instagram Story Poll after a complete 24-hour window.

Collect eight separate Story Polls before the event. Post them at the same time, use the same Poll format, keep each live for 24 hours, and do not boost or selectively reshare one Story. Use the common LIB/BOOK/SCM formula.

Required fields for each observation:

Field	Rule
Story number	1-8
Posted at	Same scheduled time each day
Measured at	Exactly 24 hours after posting
Total votes	Aggregate count only
Evidence	Admin screenshot or export reference
Approval	Checked before the price schedule is locked

Do not use one post's day-by-day new likes because the launch day normally has the highest activity. Do not use live Math Week voting because players could influence the stock. Never store voter names, Instagram handles, or responses at person level.

SCM status: Pending data collection. No SCM price path should be published until all eight approved observations are available.

Money-only leaderboard

portfolio_value = cash + sum(units_held x latest_price)

- Rank eligible players from highest to lowest portfolio value.
- A player who never trades remains at M$5,000.00.
- Equal values to two decimal places share the same rank unless a separate tie rule is approved before launch.
- The public leaderboard shows only rank, public handle, and portfolio value.
- Email, student ID, real name, individual borrowing data, donor identity, Story voter identity, and detailed holdings remain private.

Administrator lock and daily operation

Before launch

1. Confirm the four source definitions and the selected periods with the Library.
2. Obtain all eight SCM observations.
3. Calculate every return and price in advance.
4. Conduct a second-person check against the source files.
5. Lock a versioned data-and-price schedule. Record who approved it and when.
6. If a fifth index is required, approve its source and formula separately before launch.

Each game day

1. Open trading at the fixed price.
2. Let each player make at most one action per index.
3. At the cutoff, stop new orders.
4. Reveal only the next locked observation and price.
5. Recalculate portfolio values and the provisional leaderboard.
6. Never edit an observation or price silently; corrections require an audit reason.

Final settlement

1. Close trading after the seventh price change.
2. Recompute every portfolio from the transaction log and final prices.
3. Resolve flagged accounts before prize confirmation.
4. Freeze the final money-only leaderboard.

MVP acceptance checklist

☐ Current scope is shown honestly as four confirmed indices; the fifth index is either approved or marked TBC.
☐ Every active index has eight approved observations and seven calculated price changes.
☐ Every index starts at M$100.00; every ranked player starts at M$5,000.00.
☐ The rounded one-decimal return produces the saved two-decimal price exactly.
☐ Missing or incomparable data holds the previous price.
☐ Players cannot change index prices through trading or social-media interaction.
☐ Trading is whole-unit, long-only, fee-free for the simplified MVP, with one action per index per day.
☐ The leaderboard uses portfolio value only; no points or bonuses exist.
☐ Public output contains no personal or row-level Library/social-media data.
☐ Final rankings can be reconstructed from locked prices and the transaction log.

Sources

- (2023-2025) XMUM Library Gate Count.xlsx - 2025.csv
- Circulation transaction report 2024-2026 - Sheet1.csv
- DONA_dataset(Sheet1).csv
- SCM Story Poll observations: pending collection and approval

Related

- MAT_MathWeek_2026_Stock_Market_and_Leaderboard_Design_Spec.md
- MAT_MathWeek_2026_Stock_Market_and_Leaderboard_Design_Spec_ZH.md